Questions remain around cleanup of former nuke sites near Cheyenne

By Trevor Brown | Wyoming Tribune Eagle | Dec. 13, 2015

CHEYENNE - Leo Garcia hadn't heard of trichloroethylene about four decades ago when he moved into his red-and-white house just west of Cheyenne.

He didn't know the colorless chemical, also known as TCE, has been linked to liver cancer and several other health problems.

And he didn't know that he would end up living near the edge of a slow-moving contaminated groundwater plume that is troubling him and many others in Laramie County.

"There is definitely concern that they don't know where the pollution exactly is," he said. "We know it's getting near the city, and so living here on the outskirts, I think we have some legitimate concerns."

Garcia's property on Southwest Drive lies just to the east of the Interstate-80/I-25 interchange.

About 12 miles to the west of his house, tucked away in the Belvoir Ranch, is the former Atlas Missile Site 4.

That is where the Army Corps of Engineers plans to spend tens of millions of dollars on a more-than-100-year effort to remove TCE from the groundwater after it seeped into the ground while being used to clean the missile launcher in the 1960s.

And this is just one of 13 former F.E. Warren Air Force Base sites that have similar problem.

The Army Corps says there are currently no public safety risks, and the federal agency has repeatedly stated it is committed to the long and costly remediation work at the sites.

But the work has drawn an array of criticism from local residents, elected officials and former and current state regulators who question the speed and effectiveness of the clean-up plans.

And the problem is especially concerning for those, like Garcia, who live near the contamination.

So far, tests have shown that Garcia's water is safe and he and his wife still routinely drink it.

But, with the potential of the contamination spreading, he said he worries about the future.

"I don't how much longer I'm going to be around," said the 77-year-old. "But hopefully someday my children will live here, and I know that this can definitely affect them"

Standing 75 feet tall, the Atlas D missile was the first intercontinental ballistic missile deployed on American soil.

Capable of flying 5,500 miles and packed with a 1.44 megaton nuclear warhead, the weapon stood on alert in locations throughout Wyoming, Colorado and Nebraska during the tense Cold War era.

Through the years, the Air Force has taken great care to make sure there was no accidental radiation exposure at its launch sites

But, unknown to the missile maintenance officers at the time, a common cleaning practice would unleash a different dangerous carcinogen into environment.

"The presence of TCE is attributed to its use to clean the rocket engines and possibly the fuel tank of impurities after readiness exercises," a 2002 Army Corps report reads. "This may have also been performed to remove any impurities from liquid oxygen lines to prevent an accidental explosion."

The readiness exercise occurred "a few times per week," according to the report. But officials don't know exactly how much of the chemical seeped into the soil at the various missile sites.

Depending on the site and its underground geology, the contamination has spread to about the size of a football field or, like at Missile Site 4, a massive 12-mile-by-3-mile plume.

"Generally speaking, they are significant in size," said Kevin Frederick, administrator of the water quality division of the Wyoming Department of Environmental Quality. "TCE is very mobile in groundwater, and obviously a lot of it was used and released into the environment.

"This is not your grandma's leaking underground storage tank site or a landfill, for that matter."

And even though the missile sites were shut down decades ago, their impact will remain much longer.

"They operated between 1960 and 1965," said Drew Reckmeyer, a section chief with the Corps' Environmental Remediation Branch. "It's a very short period of time when you think here we are 50 years later talking about this and some of these missiles operated for only five years."

It's approaching 9 p.m. on a recent Thursday.

About two dozen people are sitting around a square-shaped table in a dimly lit classroom inside Laramie County Community College's Health Sciences Building.

The group is following a PowerPoint presentation as they go through slide after slide of the latest water-sample results.

They are state regulators, city water managers, geological experts and local residents - some of whom have environmental or scientific background and others who are just worried about the future of the area's water supply.

Together, they make up the Restoration Advisory Board for Missile Site 4. The group was formed in 2011 to help advise the Army Corps on the project. The board usually meets every three months to go over what areas have been tested, what the latest water samples show and what's next for the project.

The mix of backgrounds and personalities can lead to tense sessions, with local residents firing questions and sometimes accusations at the Army Corps representatives who traveled from their headquarters in Omaha, Nebraska, for the meeting.

But Reckmeyer explained that the federal government is committed to cleaning up these sites - and many other - no matter how long it takes.

The problem, he says, is that there is limited available funding and thousands of other sites across the country dealing with similar issues.

In June 2014, the Department of Defense reported to Congress that it had 38,804 current or former defense sites that have been contaminated with hazardous substances or pollutants.

"It's a very expensive clean up activity, and you are competing for funds," Reckmeyer said. "There are only so much funds for the (Formerly Used Defense Site) program a year."

The U.S. Department of Defense has spent billions on clean up and restoration work at these installations over the past three decades.

From 1986 to 2008, the department spent nearly $30 billion. And in fiscal year 2013 alone the department spent about $1.8 billion.

The department additionally reported last year that it has $58.6 billion - enough money to run Wyoming's state government for about 17 years - worth of work still to go. The FUDS budget for fiscal year 2016, meanwhile, is only $206 million.

The costs for some of the local sites have also ran into the millions without a clear end in sight.

Working on the missile-site contamination issues was one of the first assignments for Jane Francis when she joined the Wyoming Department of Environmental Quality in early 2001.

"The day I was hired, I met with the Cheyenne Board of Public Utilities because they were detecting TCE," she said.

More than 14 years later, and now retired from her position as a geological supervisor with the state agency, Francis hasn't stopped working on the clean-up effort.

She is now representing the community as a member of the Restoration Advisory Board for Missile Site 4.

"I guess you could say this is kind of my life's work," she said. "I have 14 years at DEQ and 15 years in training before this project, and I just think we can be doing a better job than we are doing."

Francis, more than most, understands the extent of the pollution at Missile Site 4 and the complexity in trying to monitor, contain and clean it up.

"(The Army Corps) has done some good work," she said. "But we've been working on Missile Site 4 for 14 years now, and if we knew how bad it was at the beginning, maybe we would've had our arms around it by now."

The calls for quicker action are nothing new either.

In the late 2000s, the Army Corps was hesitant to take responsibility for the entire contamination plume that was discovered east of Missile Site 4.

Frustrations that the federal agency wasn't doing enough climbed all the way up to U.S. Sen. John Barrasso, R-Wyoming.

In a 2008 speech on the Senate floor, he leaned on the Army Corps to take greater responsibility.

"The city of Cheyenne should not have to shoulder the entire burden of cleaning up this mess," he said at the time. "I believe the federal government has a responsibility to leave Wyoming the way they found it: unspoiled."

The Army Corps ended up following his recommendation and would later propose a clean-up method at the site.

The idea was to install a groundwater extraction system that would "intercept" the contaminant as it moves downstream. The extracted groundwater would then be treated and sent back to the aquifer.

Highlighting the difficulties in getting a project going, the state and others have raised concerns that this method might not work.

That led to the Army Corps scrapping the plans - at least for now - while it tries to collect more data to better understand the site.

Reckmeyer said it will probably be another year or two before a new solution is proposed.

But that delay is again prompting calls that there is a lack of urgency to get something underway.

Rep. James Byrd, D-Cheyenne, said the Legislature is even planning to get more involved to make sure the Corps' understand the importance of the project.

"This now goes through the state legislative branch in addition to the DEQ," he said. "So you know we've broaden the scope and brought some new faces to the table."

He added that the issue is becoming more timely as he and others are working on a plan to develop the city-owned Belvoir Ranch.

There is another challenge for the Army Corps: Each missile site is unique.

The underground geology, amount of chemicals that was released and type of launcher differ for each site.

For example, there is a relatively small plume of contaminated groundwater at Missile Site 6, which is near the small Laramie County town of Meriden.

The TCE-tainted water has been contained to about a 500-feet-long by 800-feet-wide area.

The Corps has spent $1.8 million on the site so far. And preliminary estimates show only $558,000 is needed to close out the site.

That is a comparatively easy fix compared to efforts to clean up Missile Site 4.

The Corps has spent $20.2 million so far on this site. And the initial projections, which federal officials said could easily change, peg the remaining cost at more than $75 million.

And there is no simple or standard fix for many of these sites. But there are some innovative options being proposed.

The Corps, for example, is planning an estimated $26 million federally funded "bioremediation" project that would inject microorganisms into the water to break down the TCE over a projected 200-year period at Missile Site 3.

Located near Carpenter, there are concerns the 7,800-foot plume of contaminated groundwater could continue to flow east toward several county residents' water wells.

State DEQ officials initially said they were concerned that the plan wouldn't treat the leading edge of the plume or have additional monitoring wells for the area.

But those provisions have since been added to the proposal.

Still, some have called on the Army Corps' to slow down before moving forward.

"I think work at that site is premature," Francis said. "I don't think they completely have their arms around it, and I'm not sure the technology they are proposing is the most cost-efficient one."

Issues, such as this one, illustrate the problems for all involved.

Do officials look for a speedy solution no matter the cost? Or should they take their time and make sure the solution is a sound one?

Meanwhile, local residents, like Garcia, who live near the areas continue to wait to see what happens next.

"It seems like it has just been the status quo so far," he said. "But I'm curious to see what type of remediation they can present."

Information from the Environmental Protection Agency Reporting by Trevor Brown, Digital graphic and data entry by Josh Rhoten

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